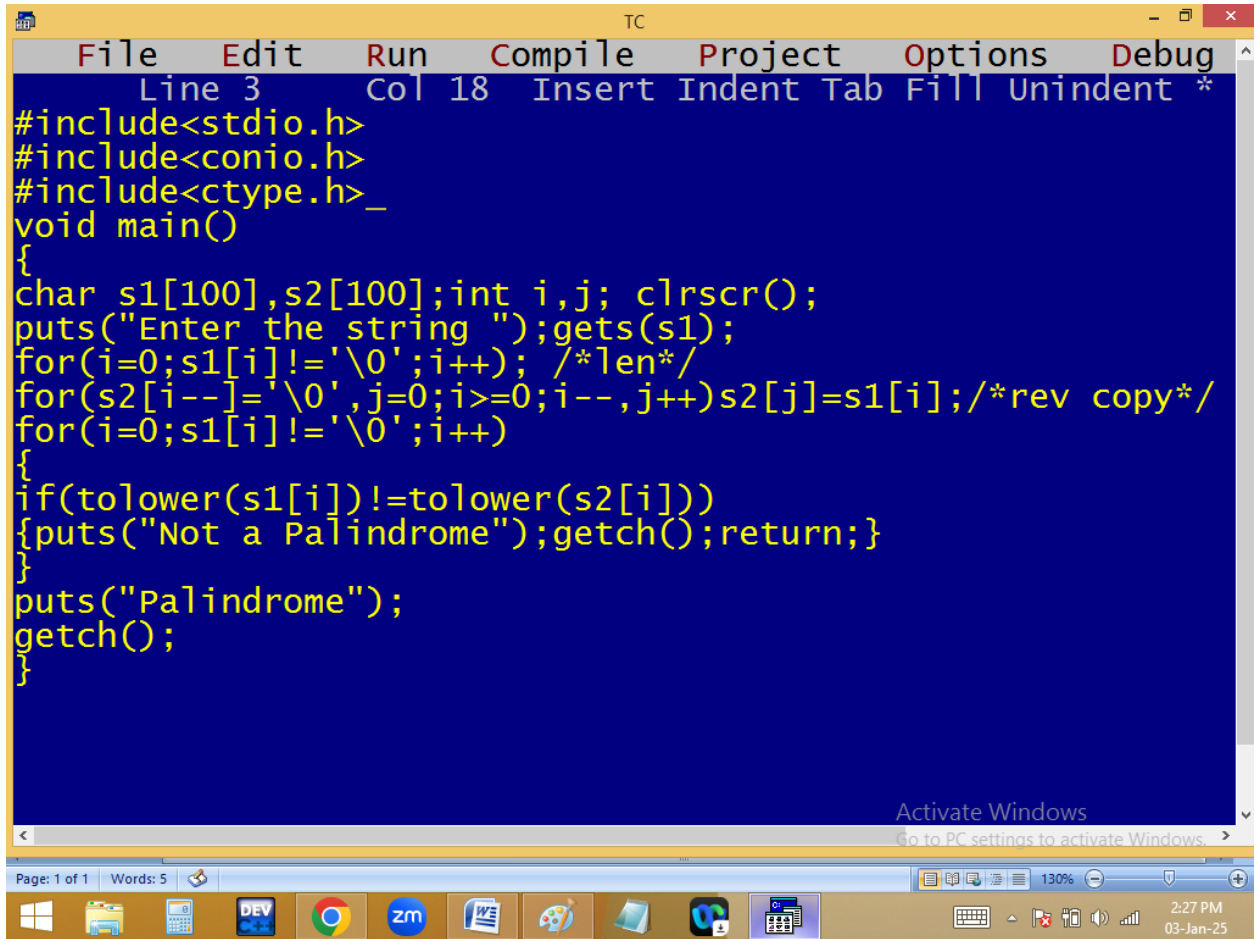


Finding palindrome using 2 strings:

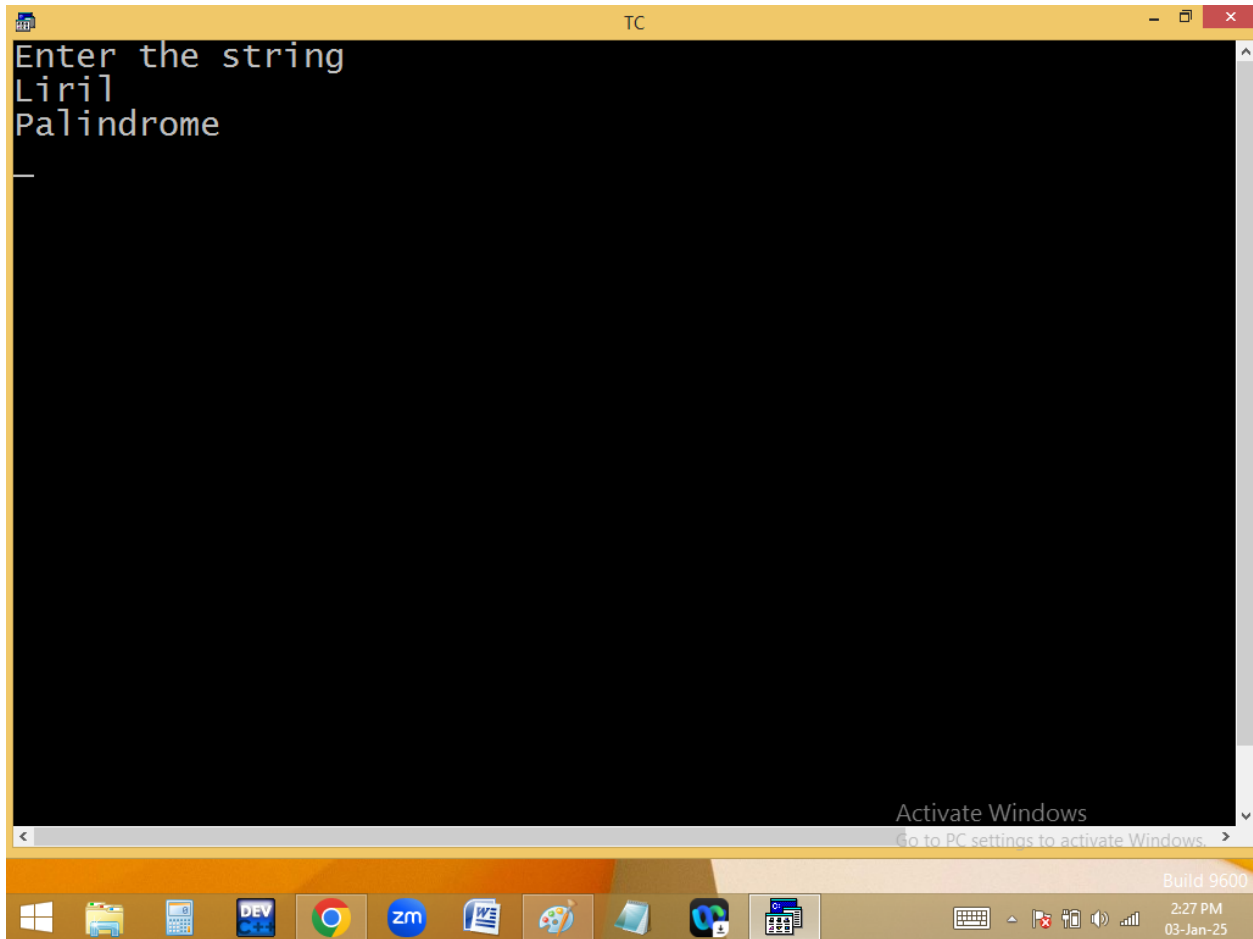


```
TC
File Edit Run Compile Project Options Debug
Line 3 Col 18 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<ctype.h>_
void main()
{
char s1[100],s2[100];int i,j; clrscr();
puts("Enter the string ");gets(s1);
for(i=0;s1[i]!='\0';i++); /*len*/
for(s2[i--]='\0',j=0;i>=0;i--,j++)s2[j]=s1[i];/*rev copy*/
for(i=0;s1[i]!='\0';i++)
{
if(tolower(s1[i])!=tolower(s2[i]))
{puts("Not a Palindrome");getch();return;}
}
puts("Palindrome");
getch();
}
```

Page: 1 of 1 Words: 5

130%

2:27 PM
03-Jan-25



```

Enter the string
loril
Not a Palindrome

```

TC

Activate Windows
Go to PC settings to activate Windows.

Page: 2 of 2 Words: 5

2:27 PM
03-Jan-25

for(i=0; s1[i]!='\0'; i++); len=4 = i

for(s2[i--]='\0', j=0; i>=0; i--, j++) s2[j]=s1[i];

str rev + copy

for(i=0; s1[i]!='\0'; i++)

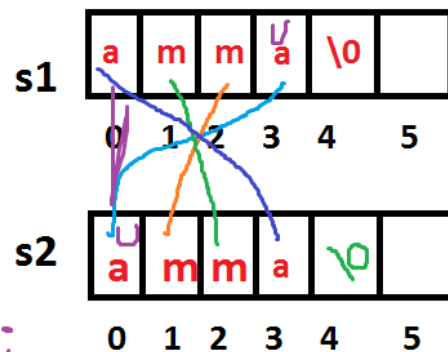
{ strcmp

if(s1[i]!=s2[i])p(not);return;

}

p(palin);

4
 3 → 0
 2 → 1
 1 → 2
 0 → 3
 -1 → 4

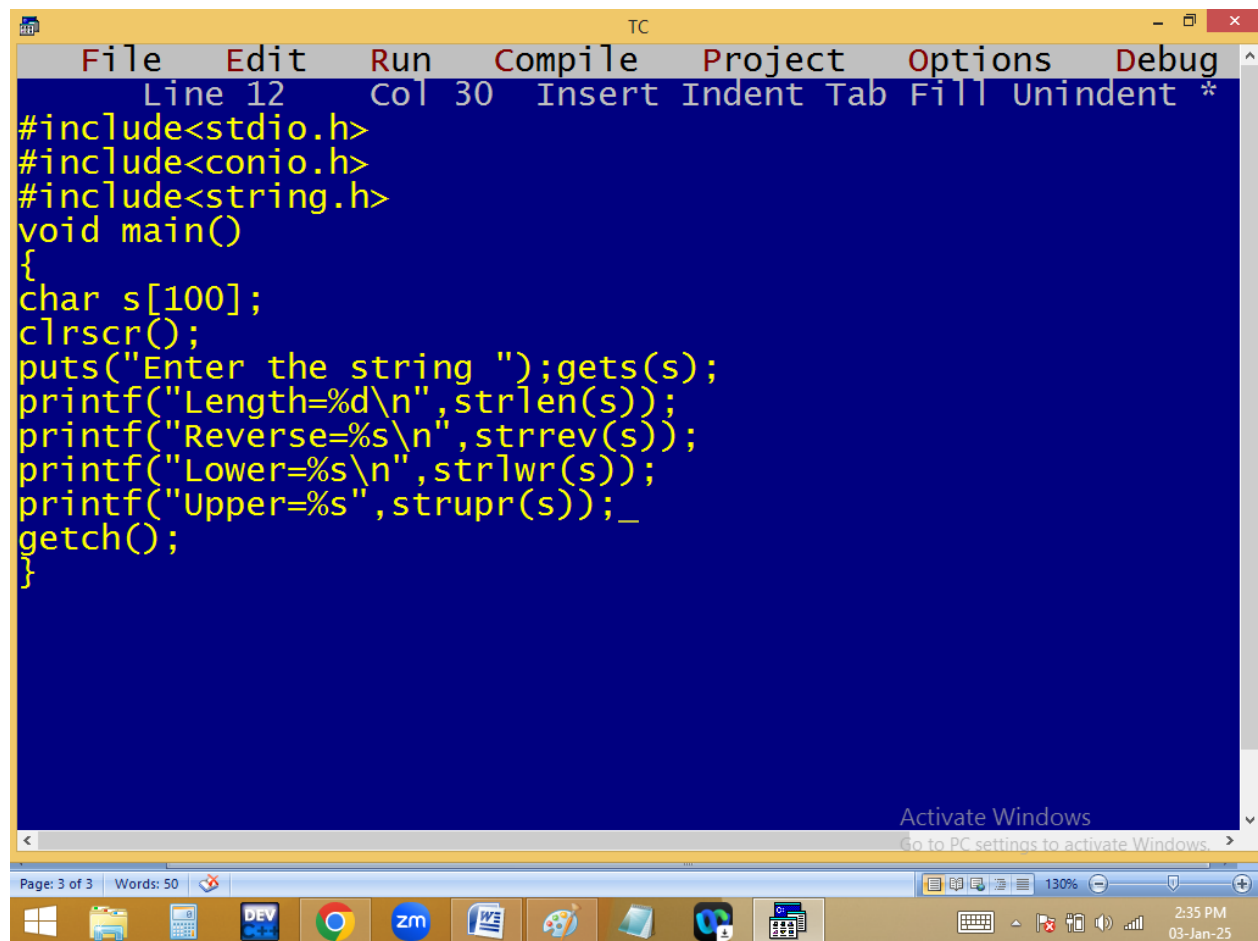


i
0 1 2 3 4

STRING LIBRARY FUNCTIONS

To make string operations easy, the c compiler provides some string predefined functions available in string.h. They are

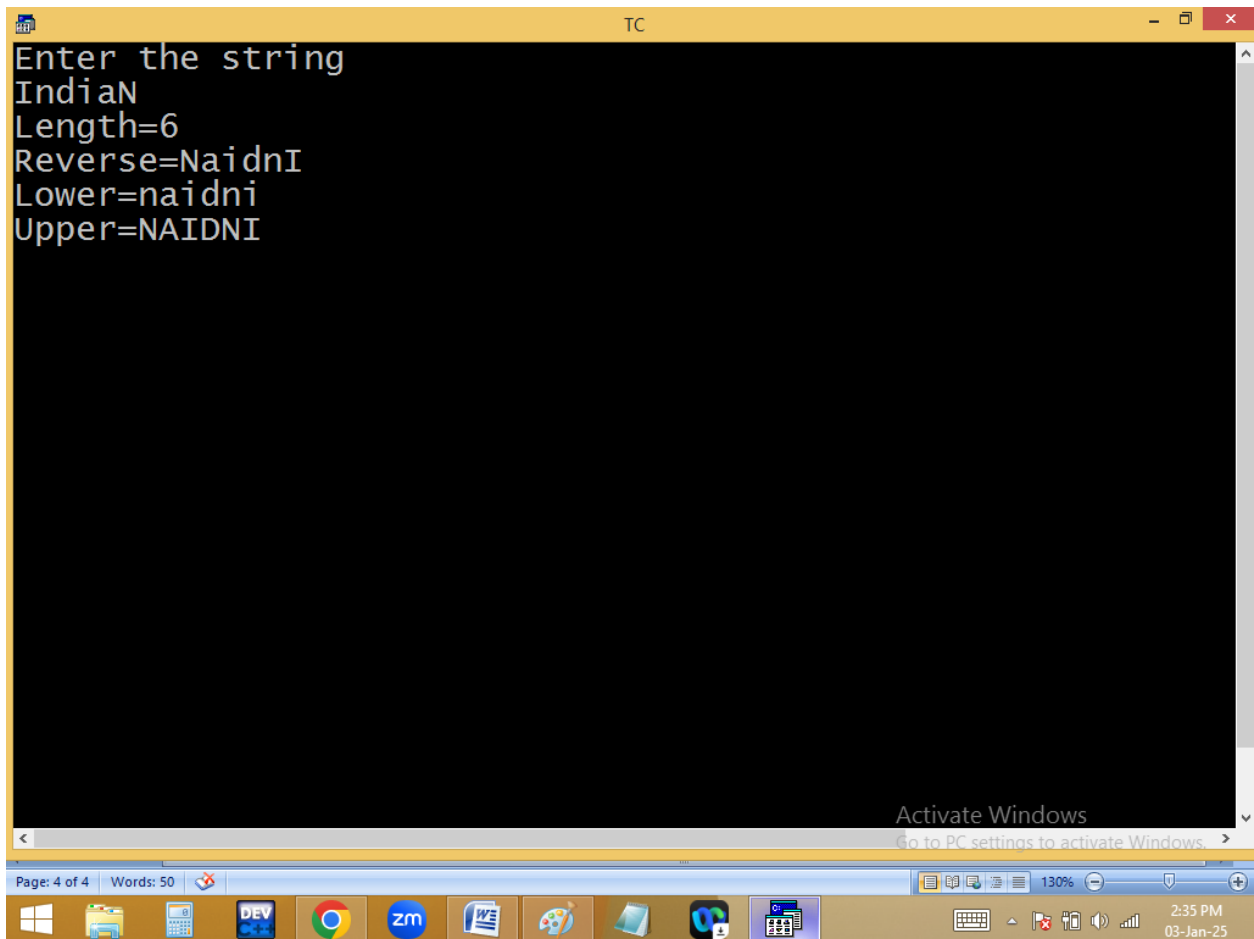
1. **strlen(): Return string length.**
2. **strrev(): Return reversed string.**
3. **strlwr(): converts string into lower case.**
4. **strupr(): converts string into upper case.**



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 12", "Col 30", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s[100];
clrscr();
puts("Enter the string ");gets(s);
printf("Length=%d\n",strlen(s));
printf("Reverse=%s\n",strrev(s));
printf("Lower=%s\n",strlwr(s));
printf("Upper=%s",strupr(s));_
getch();
}
```

At the bottom of the IDE window, there is a status bar showing "Page: 3 of 3" and "Words: 50". Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and other applications. The system tray on the right shows the time as "2:35 PM" and the date as "03-Jan-25". A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.



The screenshot shows a Turbo C++ (TC) compiler window with a black background and white text. The text displays the result of a string reversal operation on the word "Indian". The output shows the original string, its length (6), the reversed string "NaidnI", and the strings converted to lowercase ("naidni") and uppercase ("NAIDNI"). The window title bar reads "TC". At the bottom, there is a Windows taskbar with various application icons and a system clock showing 2:35 PM on 03-Jan-25.

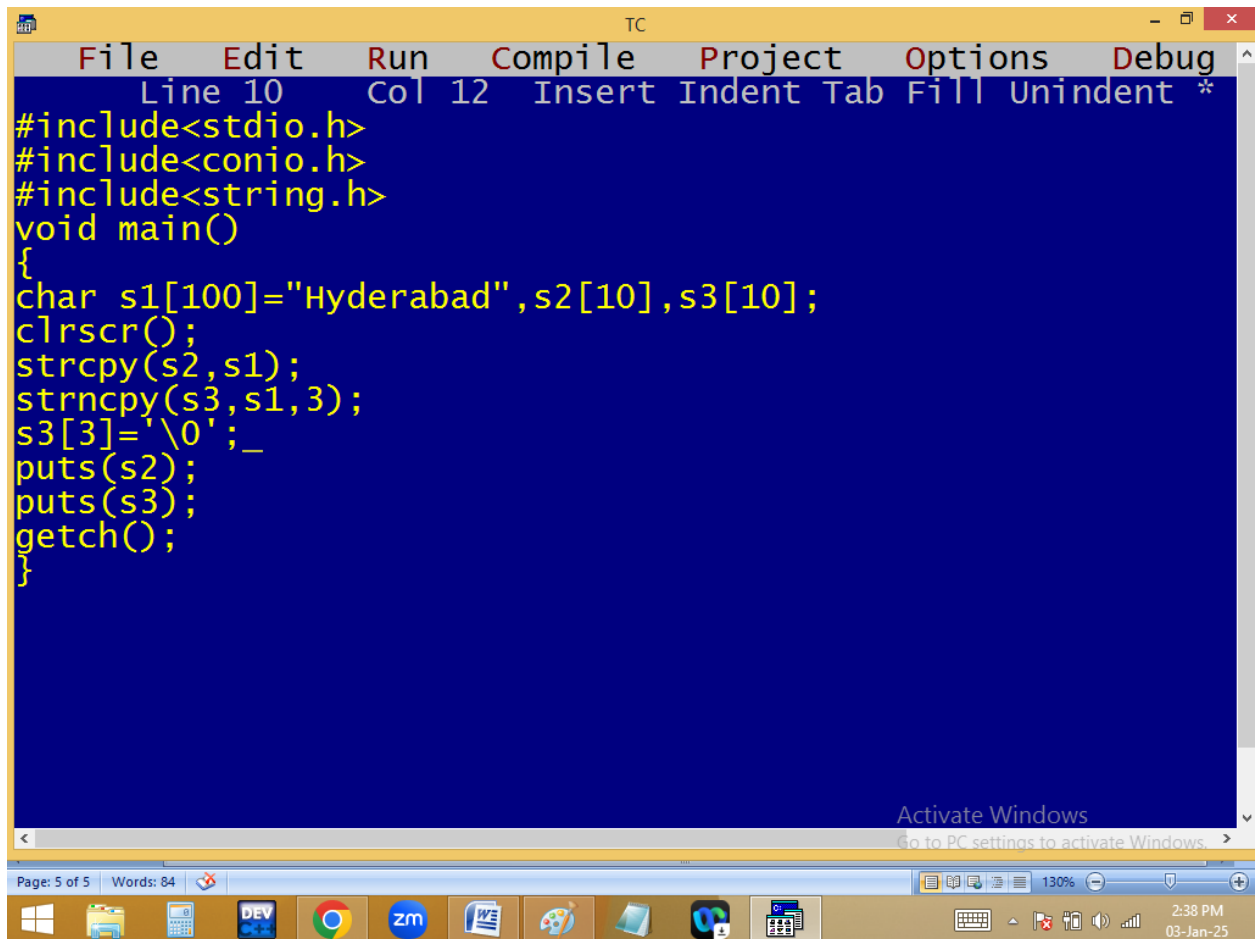
```
Enter the string
Indian
Length=6
Reverse=NaidnI
Lower=naidni
Upper=NAIDNI
```

5. strcpy(): It copies the source string into destination string.

Strcpy(destination string, source string);

6. strncpy(): It copies specified no of characters into destination string.

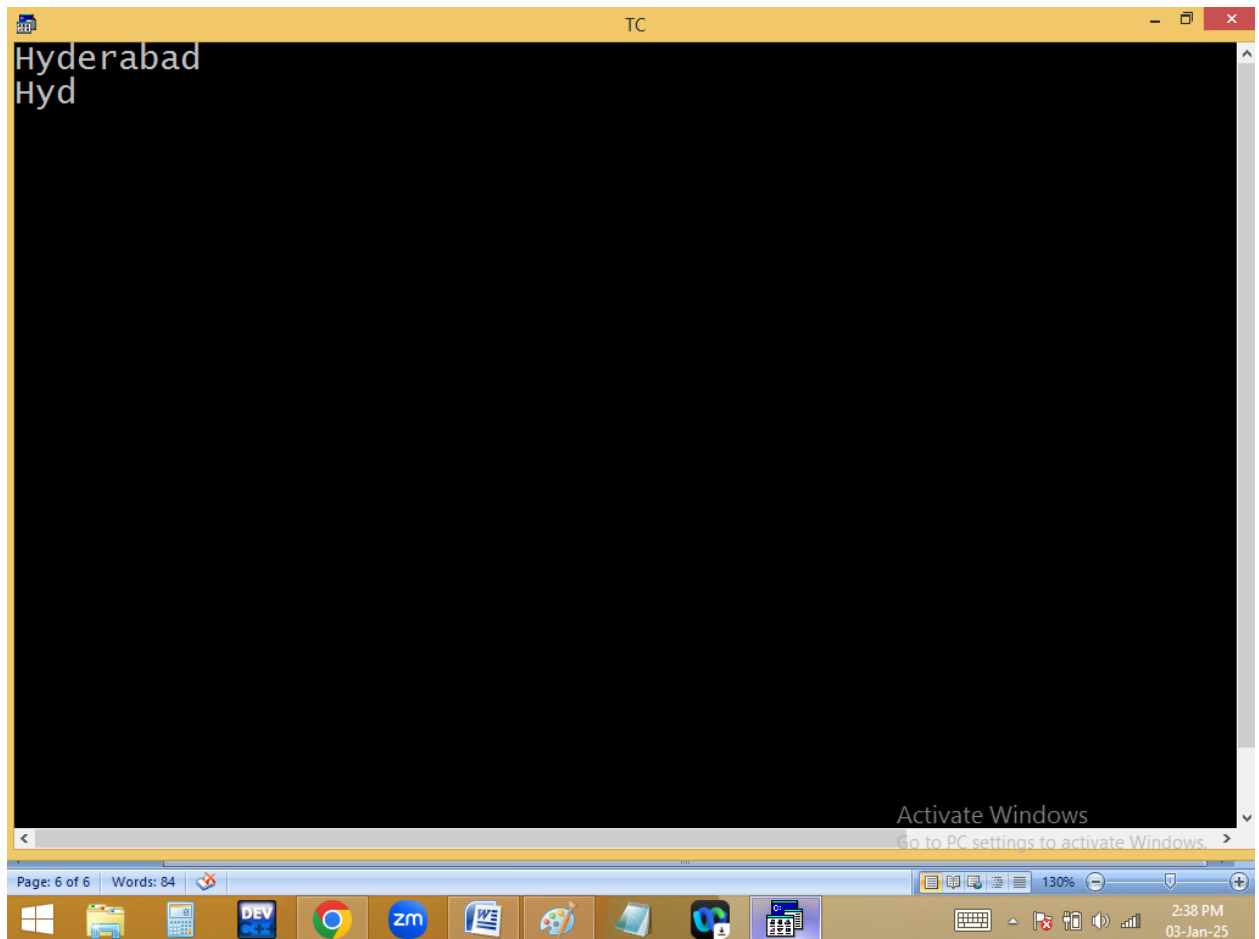
Strncpy(destination string, source string, no of char);



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 10", "Col 12", and lists editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s1[100]="Hyderabad",s2[10],s3[10];
clrscr();
strcpy(s2,s1);
strncpy(s3,s1,3);
s3[3]='\0';_
puts(s2);
puts(s3);
getch();
}
```

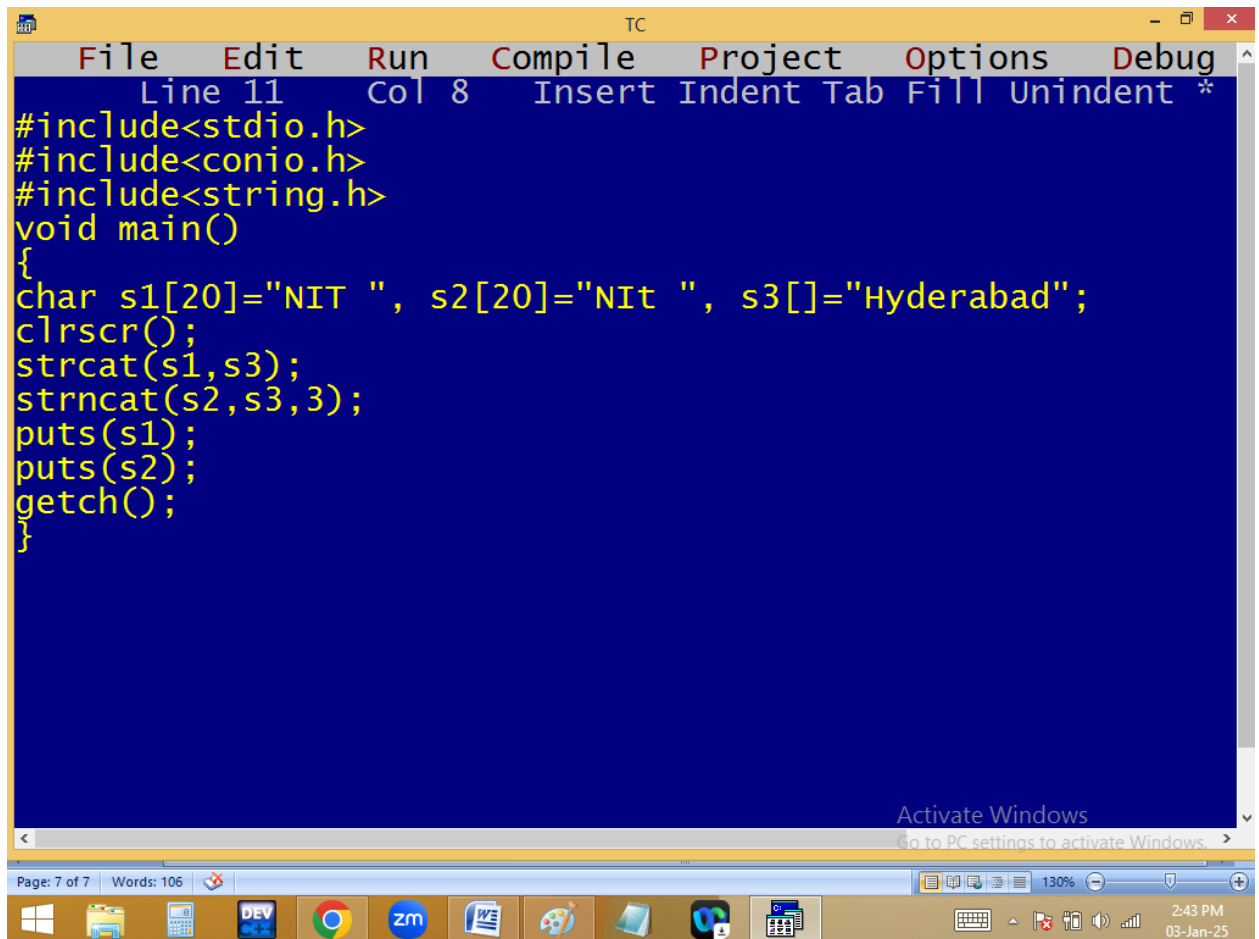
At the bottom of the IDE window, there is a status bar showing "Page: 5 of 5" and "Words: 84". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and other applications. The system tray on the right shows the time as "2:38 PM" and the date as "03-Jan-25". A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.



7. strcat(): It adds string2 with string1

8. strncat(): It adds specified no of char to string1.

Strncat(string1, string2, no of char's);



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 11", "Col 8", and provides options: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. The code is as follows:

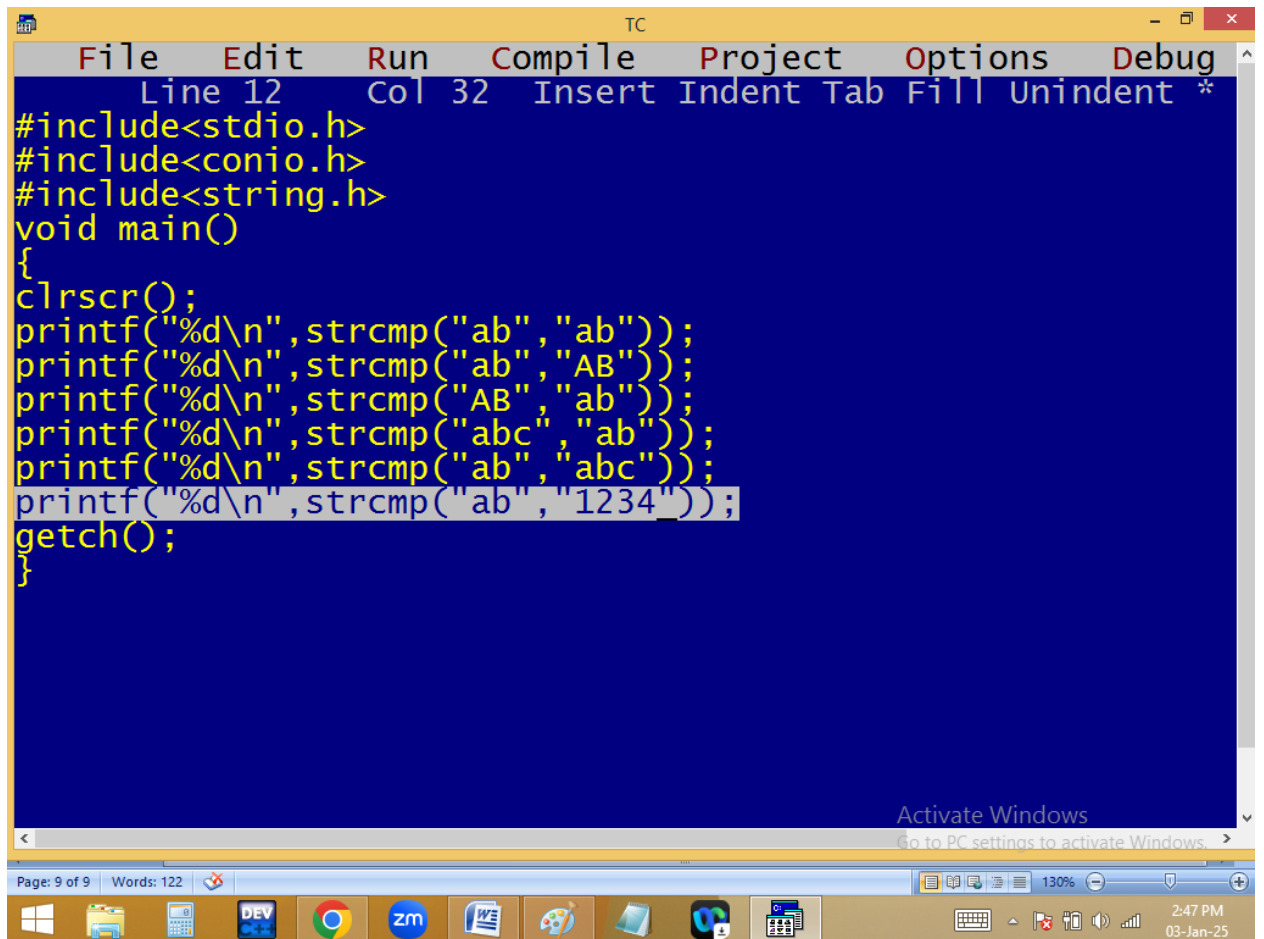
```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s1[20]="NIT ", s2[20]="NIt ", s3[]="Hyderabad";
clrscr();
strcat(s1,s3);
strncat(s2,s3,3);
puts(s1);
puts(s2);
getch();
}
```

At the bottom of the IDE window, there is a status bar showing "Page: 7 of 7" and "Words: 106". Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and other applications. The system tray on the right shows the time as "2:43 PM" and the date as "03-Jan-25".



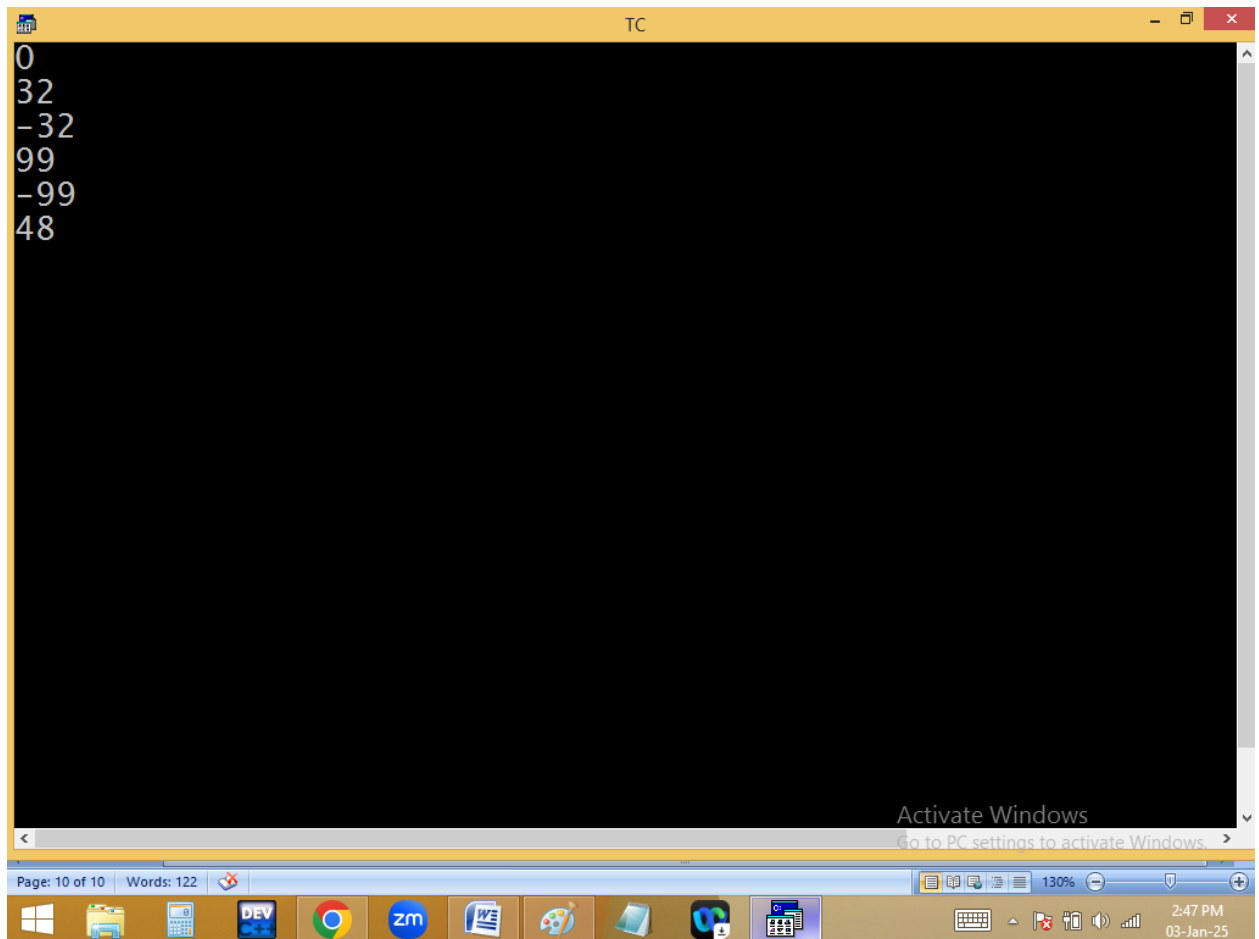
```
NIT Hyderabad  
NIt Hyd
```

9. strcmp(): It compare two strings based on ascii values and return the first ascii difference.



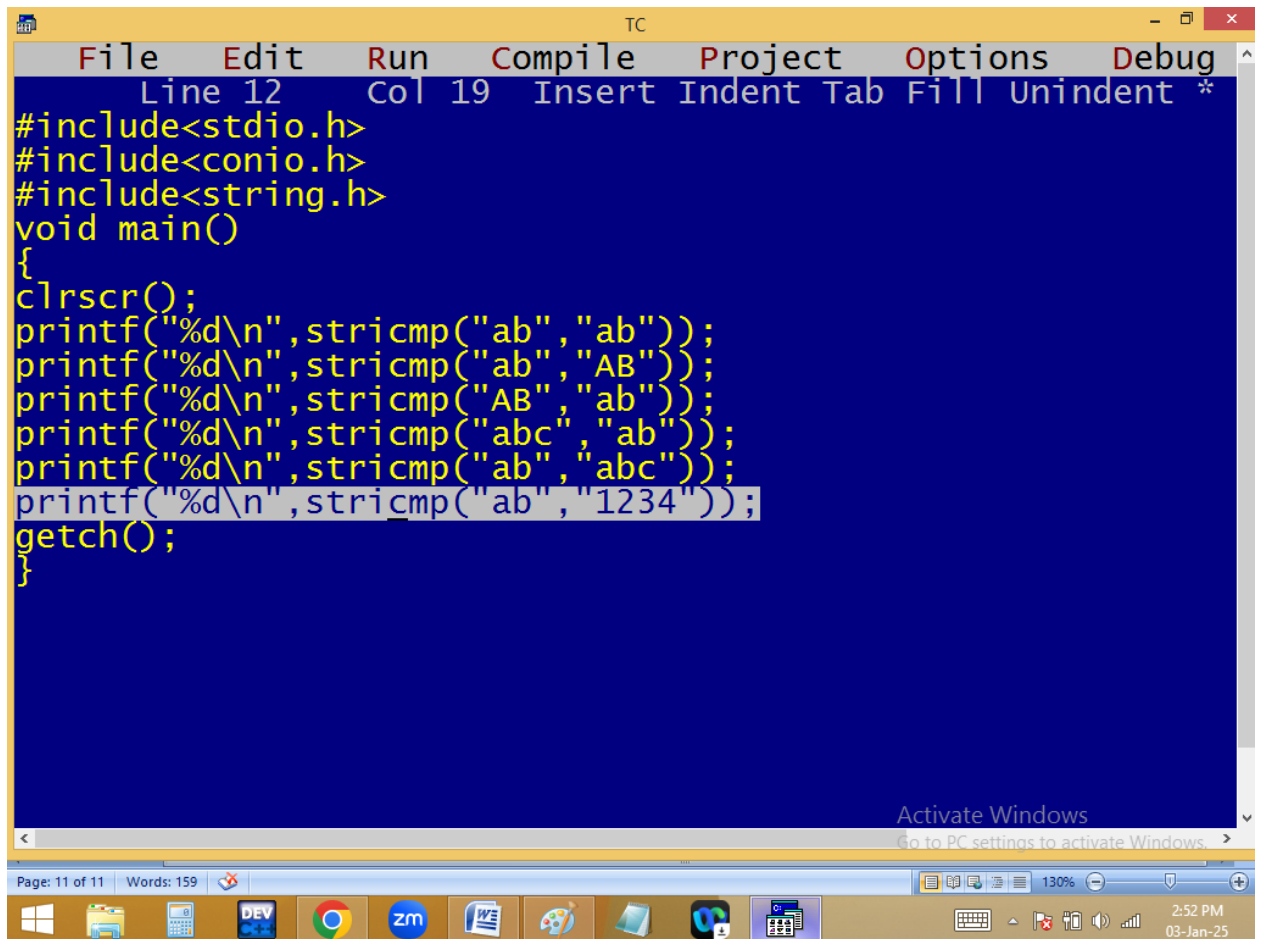
```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 32 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
clrscr();
printf("%d\n",strcmp("ab","ab"));
printf("%d\n",strcmp("ab","AB"));
printf("%d\n",strcmp("AB","ab"));
printf("%d\n",strcmp("abc","ab"));
printf("%d\n",strcmp("ab","abc"));
printf("%d\n",strcmp("ab","1234"));
getch();
}
```

Page: 9 of 9 Words: 122 130% 2:47 PM 03-Jan-25



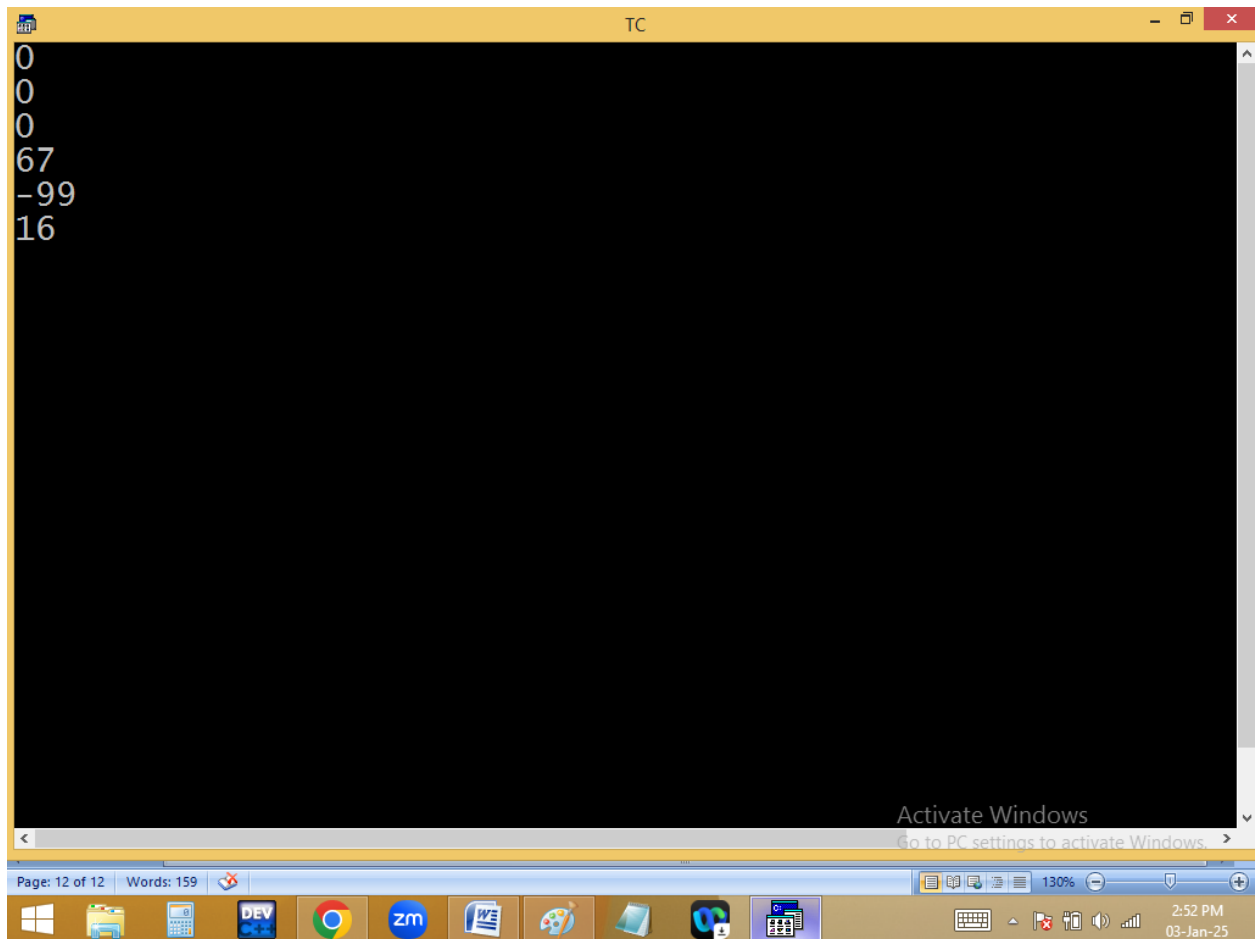
```
0
32
-32
99
-99
48
```

10. **stricmp():** It compare two strings by ignoring case. i.e. in stricmp() lower and upper are same. When different data type found or no matching char found in 2nd string, the first string converted into upper case.



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 19 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
clrscr();
printf("%d\n",strcmp("ab","ab"));
printf("%d\n",strcmp("ab","AB"));
printf("%d\n",strcmp("AB","ab"));
printf("%d\n",strcmp("abc","ab"));
printf("%d\n",strcmp("ab","abc"));
printf("%d\n",strcmp("ab","1234"));
getch();
}
```

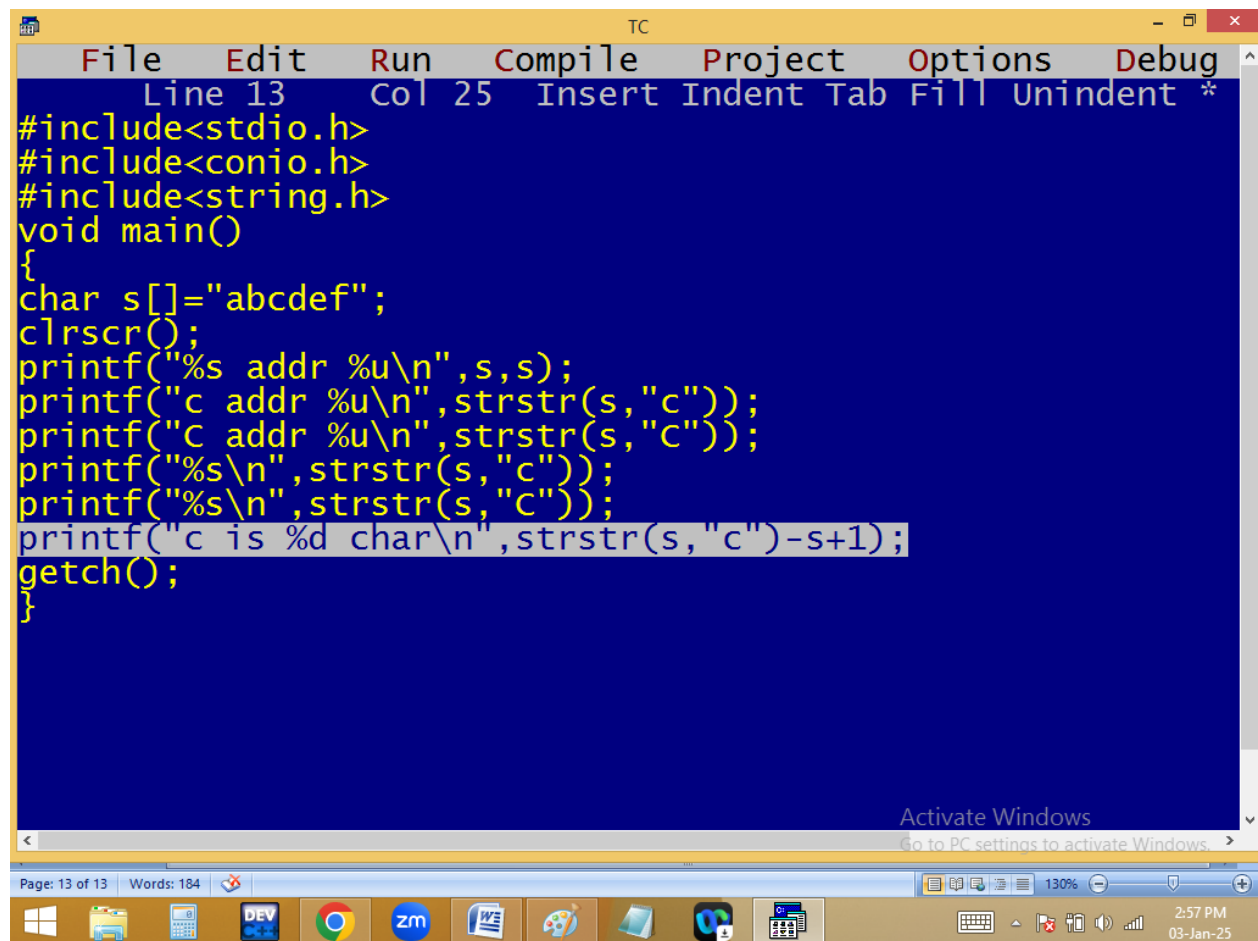
Page: 11 of 11 Words: 159 130% 2:52 PM 03-Jan-25



```
0
0
0
67
-99
16
```

11. strstr(): It return the sub string address in main string. If sub string not found. It return (null).

Strstr(main string, sub string);

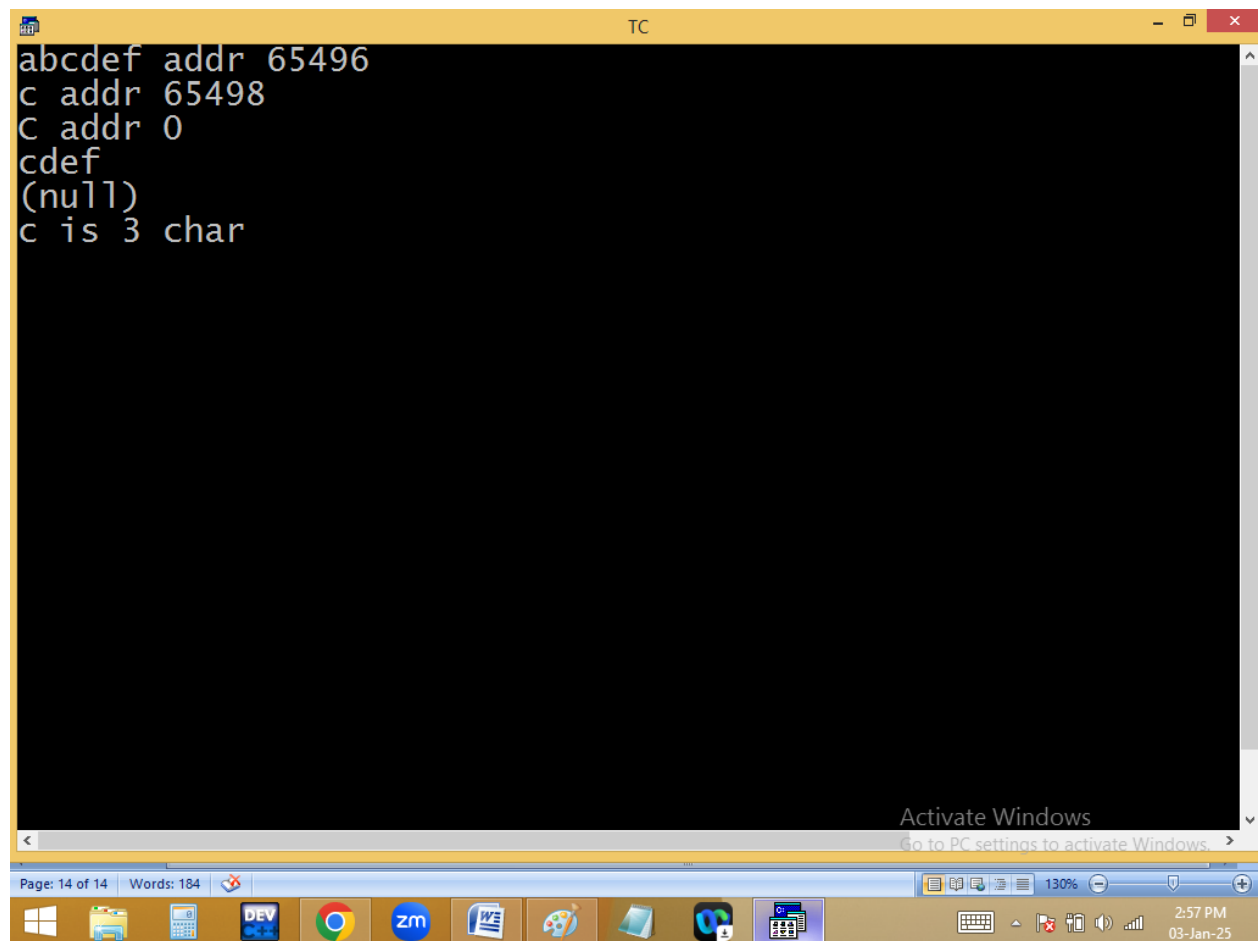


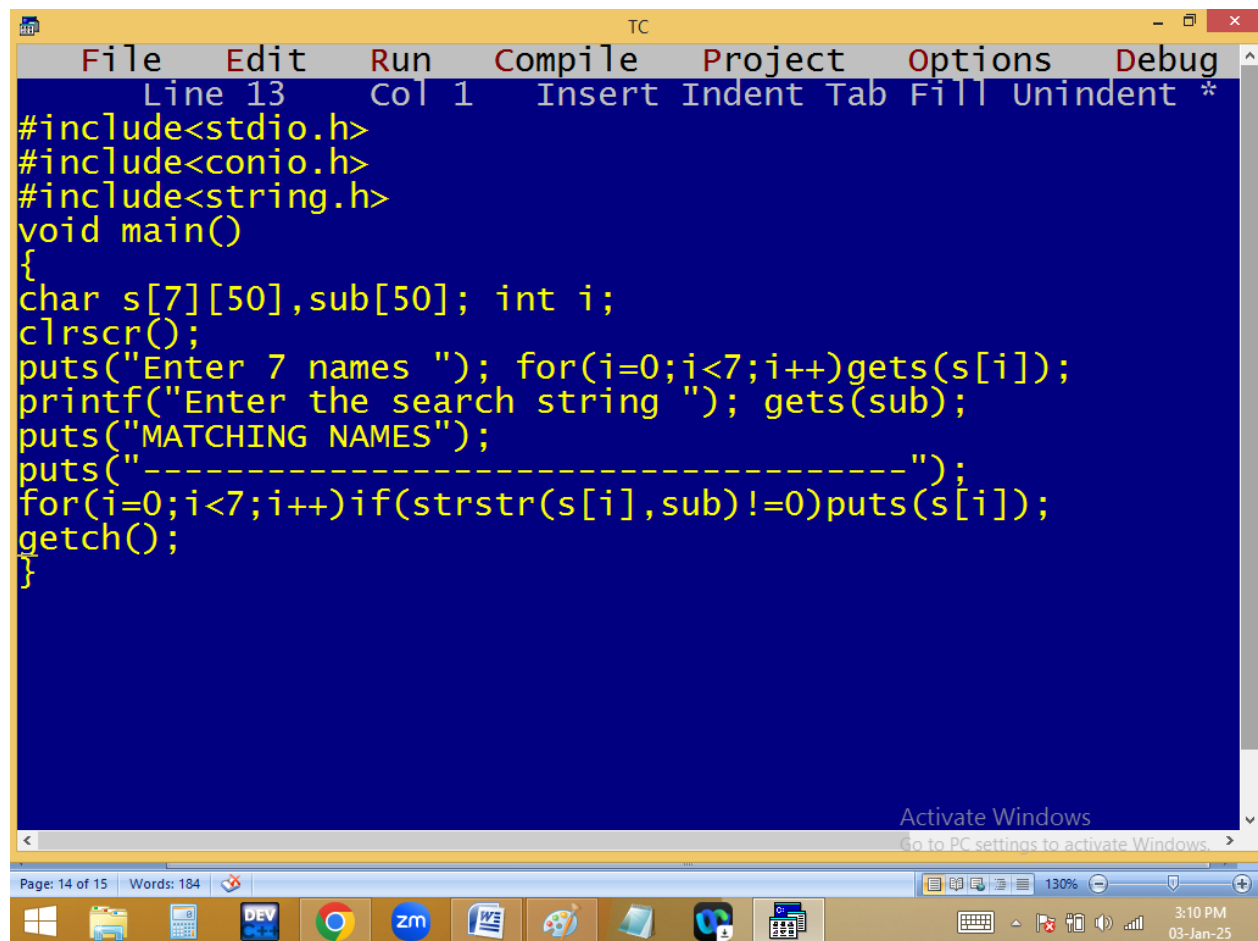
```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 25 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s[]="abcdef";
clrscr();
printf("%s addr %u\n",s,s);
printf("c addr %u\n",strstr(s,"c"));
printf("C addr %u\n",strstr(s,"c"));
printf("%s\n",strstr(s,"c"));
printf("%s\n",strstr(s,"c"));
printf("c is %d char\n",strstr(s,"c")-s+1);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

Page: 13 of 13 Words: 184 130%

2:57 PM
03-Jan-25





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 1", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and an asterisk. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s[7][50],sub[50]; int i;
clrscr();
puts("Enter 7 names "); for(i=0;i<7;i++)gets(s[i]);
printf("Enter the search string "); gets(sub);
puts("MATCHING NAMES");
puts("-----");
for(i=0;i<7;i++)if(strstr(s[i],sub)!=0)puts(s[i]);
getch();
}
```

At the bottom of the IDE window, there is a status bar with the text "Page: 14 of 15" and "Words: 184". Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and a folder. The system tray on the right shows the time "3:10 PM" and the date "03-Jan-25". A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.

```
TC
Enter 7 names
anushka sharma
virat kohli
sachin tendulkar
anjali tendulkar
rohith sharma
dhoni
sakshi
Enter the search string sh
MATCHING NAMES
-----
anushka sharma
rohith sharma
sakshi
```

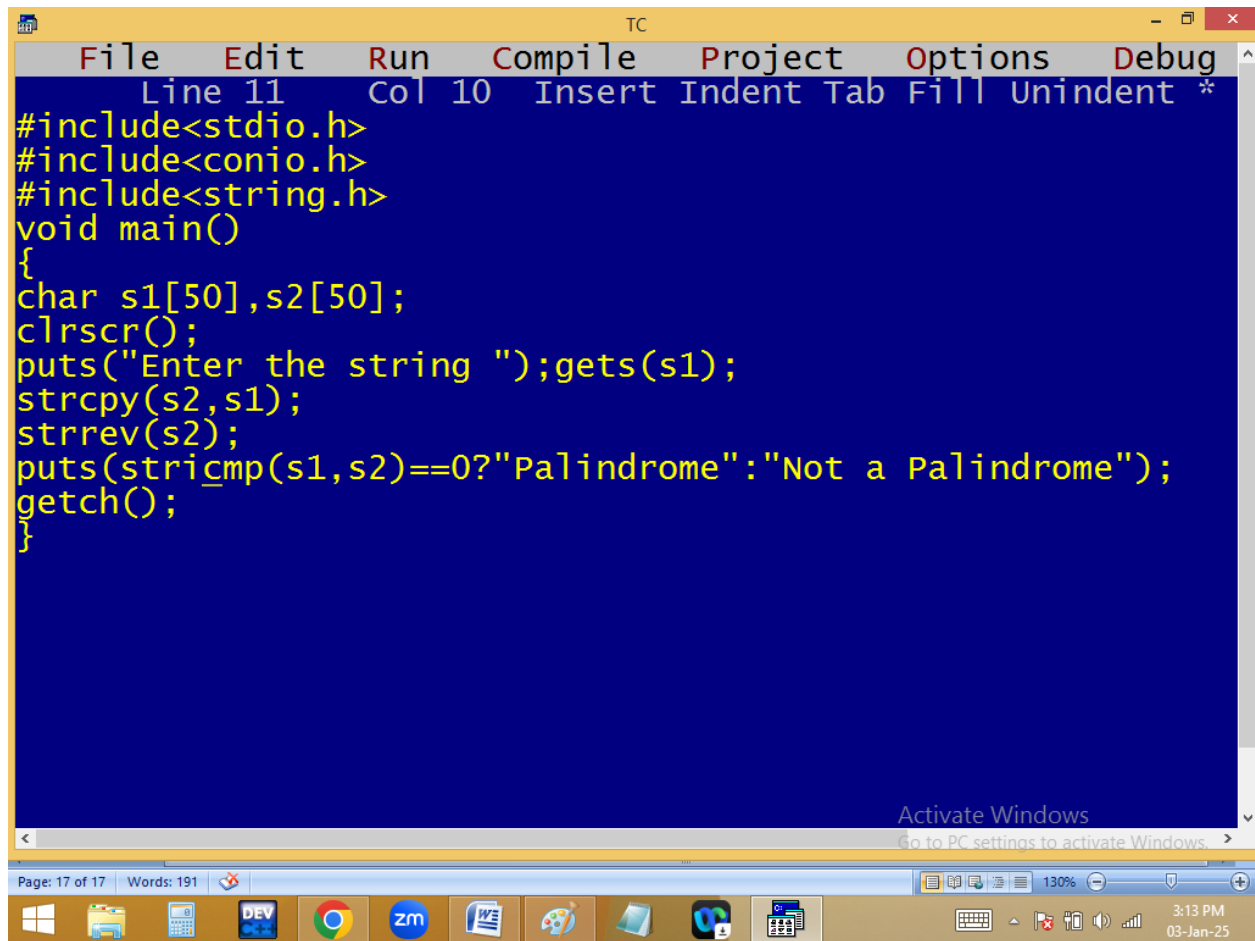
Activate Windows
Go to PC settings to activate Windows.

Page: 16 of 16 Words: 184

130%

3:11 PM
03-Jan-25

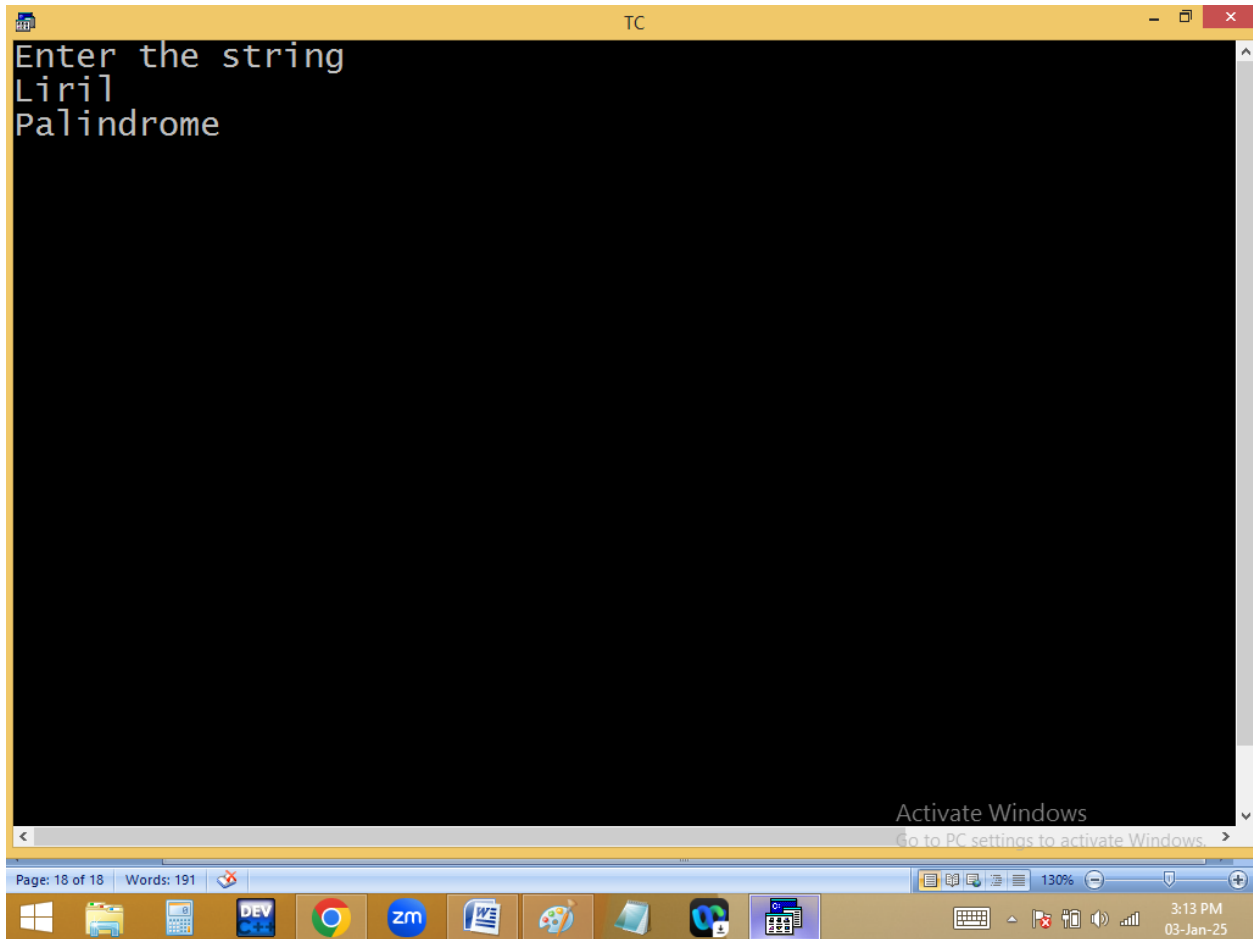
Finding palindrome or not using library functions?

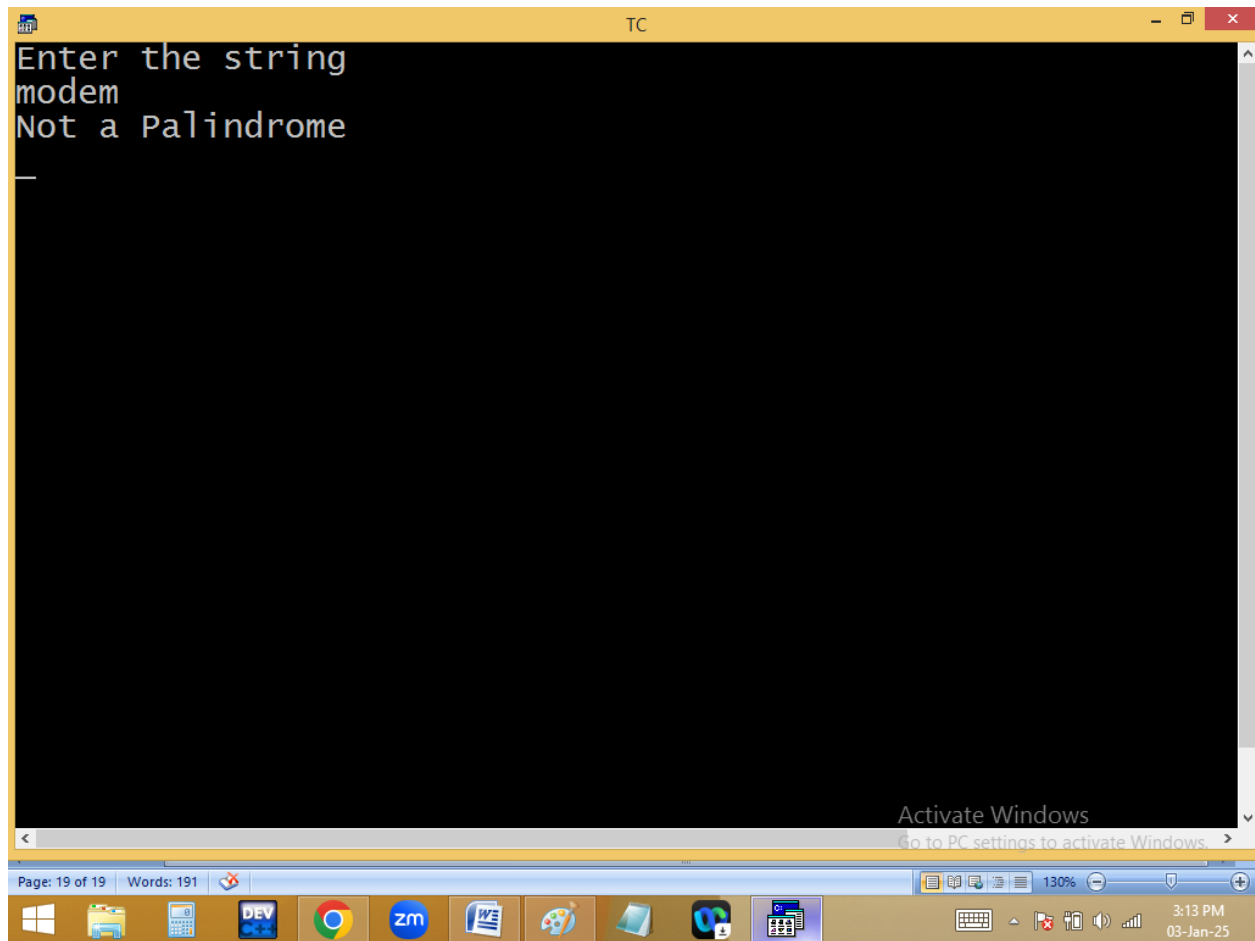


The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a blue background with yellow text. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, and Debug. Below the menu bar, a status bar shows 'Line 11 Col 10' and a list of editing actions: Insert, Indent, Tab, Fill, Unindent, and *. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s1[50],s2[50];
clrscr();
puts("Enter the string ");gets(s1);
strcpy(s2,s1);
strrev(s2);
puts(stricmp(s1,s2)==0?"Palindrome":"Not a Palindrome");
getch();
}
```

At the bottom of the IDE window, there is a status bar with 'Page: 17 of 17' and 'Words: 191'. Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and other applications. The system tray on the right shows the time as 3:13 PM and the date as 03-Jan-25.





Sorting of strings:

s[0]

~~sakshi~~ ~~meenu~~ bablu

s[1]

~~meenu~~ ~~sakshi~~ ~~pinky~~ ~~meenu~~ meena

s[2]

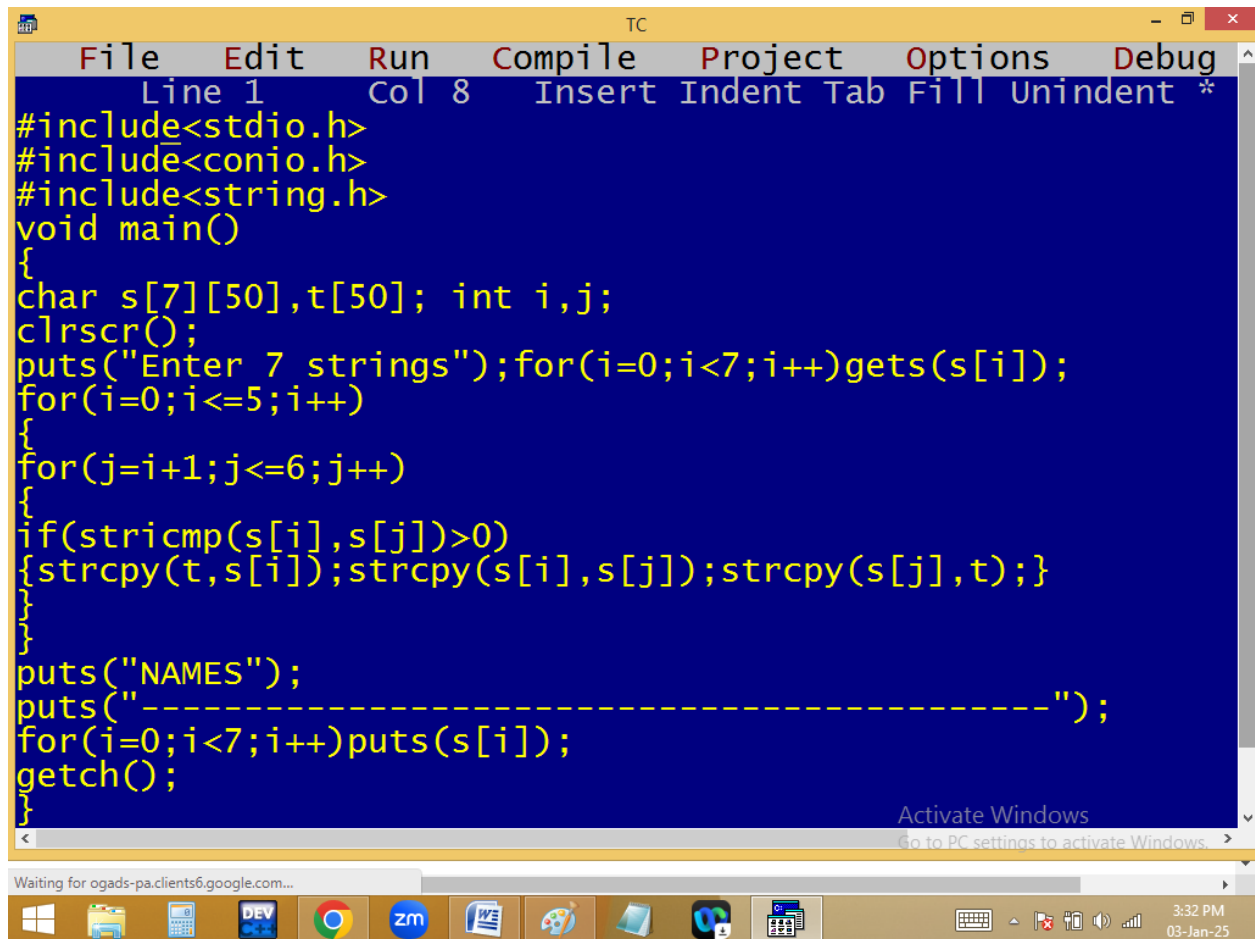
~~pinky~~ ~~sakshi~~ ~~pinky~~ meenu

s[3]

~~bablu~~ ~~meenu~~ ~~pinky~~ ~~sakshi~~ pinky

s[5]

~~meena~~ ~~meenu~~ ~~pinky~~ sakshi



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 1", "Col 8", and a list of actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and an asterisk. The main editing area has a dark blue background with yellow text. The code is a C program that sorts an array of strings using a bubble sort algorithm. The code includes headers for `stdio.h`, `conio.h`, and `string.h`. It defines a `main` function where a 7x50 character array `s` and a 50-character array `t` are declared. The program prompts the user to enter 7 strings, reads them into `s`, and then sorts them using `strcmp` and `strcpy`. Finally, it prints the sorted strings under the heading "NAMES". At the bottom of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window, a Windows taskbar is visible with various application icons and a system tray showing the time as 3:32 PM on 03-Jan-25.

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s[7][50],t[50]; int i,j;
clrscr();
puts("Enter 7 strings");for(i=0;i<7;i++)gets(s[i]);
for(i=0;i<=5;i++)
{
for(j=i+1;j<=6;j++)
{
if(strcmp(s[i],s[j])>0)
{strcpy(t,s[i]);strcpy(s[i],s[j]);strcpy(s[j],t);}
}
}
puts("NAMES");
puts("-----");
for(i=0;i<7;i++)puts(s[i]);
getch();
}
```

```
TC
Enter 7 strings
Ntr
mgr
Kishore
rekha
amithab
rajesh khanna
Sridevi
NAMES
-----
amithab
Kishore
mgr
Ntr
rajesh khanna
rekha
Sridevi
```

Activate Windows
Go to PC settings to activate Windows.

Page: 23 of 23 Words: 194 130% 3:32 PM 03-Jan-25

Character frequency:

bahu bali – 2

b- 2 time

a – 2 times

h – 1 time

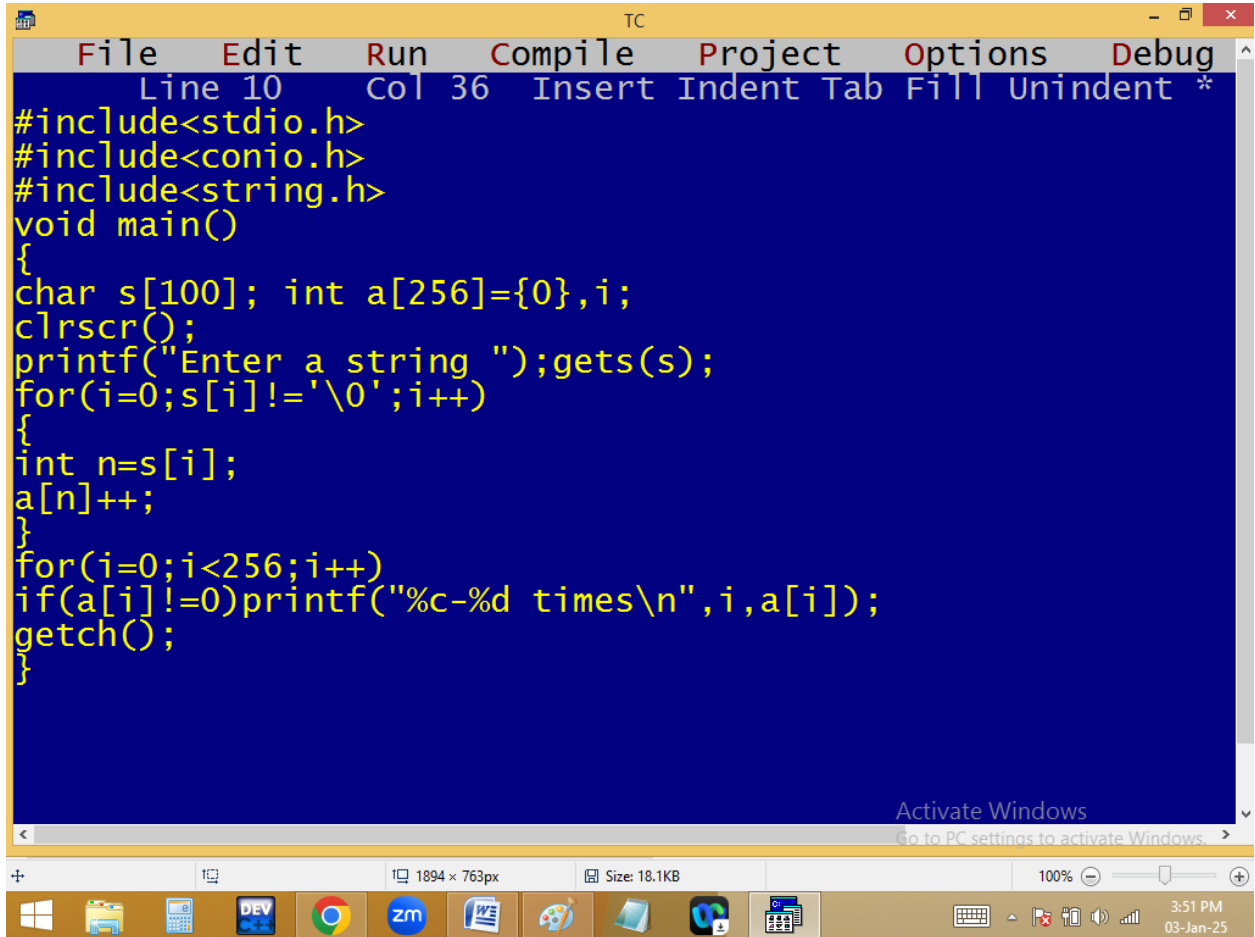
u – 1 time

l – 1 time

l – 1 time

-- 1time

2 – 1 time



The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 10 Col 36'. The main editing area has a dark blue background with yellow text. The code is a C program that counts the frequency of each character in an input string. It includes headers for stdio.h, conio.h, and string.h. The main function declares a character array 's' of size 100 and an integer array 'a' of size 256, initialized to zero. It prompts the user to enter a string and then iterates through the string, counting the occurrences of each character. Finally, it prints the frequency of each character from 'a' to 'z'.

```
File Edit Run Compile Project Options Debug
Line 10 Col 36 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s[100]; int a[256]={0},i;
clrscr();
printf("Enter a string ");gets(s);
for(i=0;s[i]!='\0';i++)
{
int n=s[i];
a[n]++;
}
for(i=0;i<256;i++)
if(a[i]!=0)printf("%c-%d times\n",i,a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

1894 x 763px Size: 18.1KB 100%

3:51 PM 03-Jan-25

```
TC
Enter a string bahu bali-2
-1 times
--1 times
2-1 times
a-2 times
b-2 times
h-1 times
i-1 times
l-1 times
u-1 times
_
```

Activate Windows
Go to PC settings to activate Windows.

Page: 25 of 26 Words: 249 130% 3:51 PM 03-Jan-25

The screenshot shows a Windows 10 desktop environment. A Turbo C++ (TC) window is open, displaying the text: "Enter a string anand", "a-2 times", "d-1 times", and "n-2 times". Below the TC window, a Microsoft Word document is open, showing "Page: 27 of 28" and "Words: 249". The taskbar at the bottom contains icons for Windows, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and the Start menu. The system tray on the right shows the time as 3:51 PM on 03-Jan-25. An "Activate Windows" watermark is visible in the bottom right corner of the TC window.

```
Enter a string anand
a-2 times
d-1 times
n-2 times
```

Home work:

I am proud to be Indian.

I ma duorp ot eb naidnl

I am proud to be Indian.

Indian be to proud am i

